Mining Scientific Papers

Knowledge Mining – Prepare for Class 4

Summary of my findings

- NLP to be used for large-scale analysis of texts and citations, as well as structures of scientific papers
- New advanced techniques (i.e., NLP4NLP, deep attentive neural network or DANN, SYMBALS, Termolator tool, etc.) can help categorizing these papers for easier access (tied to information management)
- Various challenges due to papers lacking metadata, inconsistent author identification, different formats, etc.
- Need for international collaboration to help standardize the process to make it easier for the algorithms to sort out the papers

Three questions for class discussion

- What are the international standards that everyone must follow and who gets to set them?
- When using NLP to analyze all the scientific papers, do we also take data privacy into consideration? And who is doing the analyzing?
- Can NLP help identify bias or misinformation in those scientific papers?